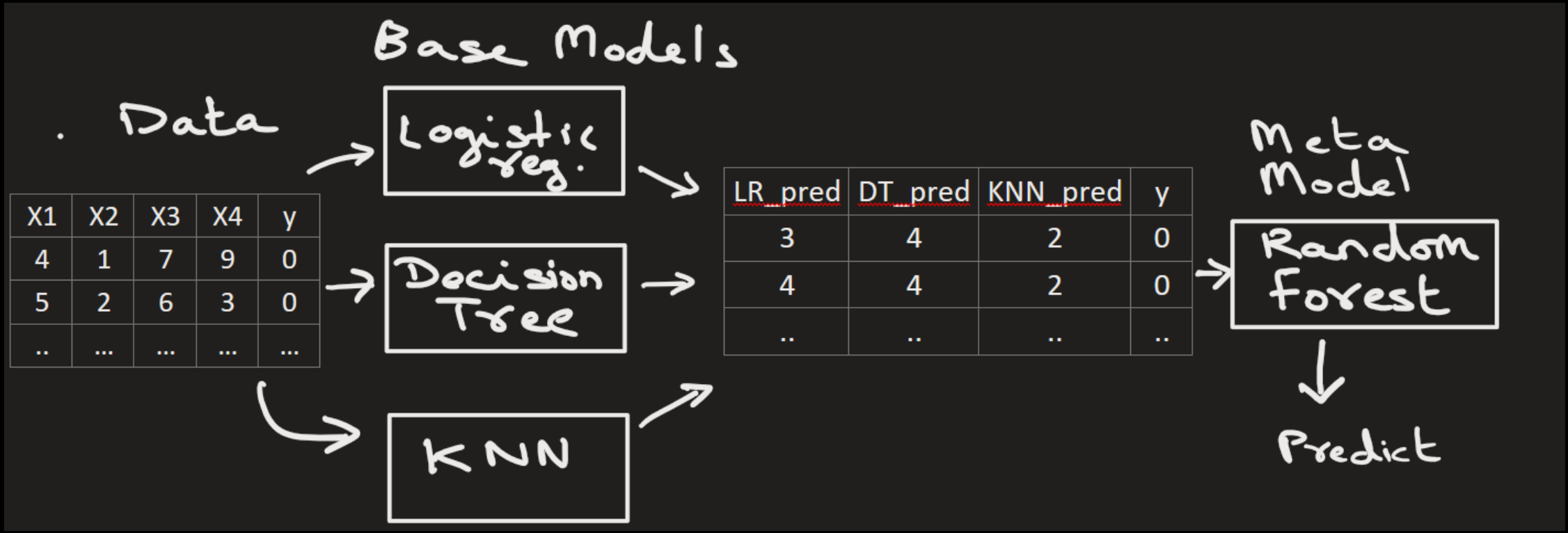
Ensemble Learning

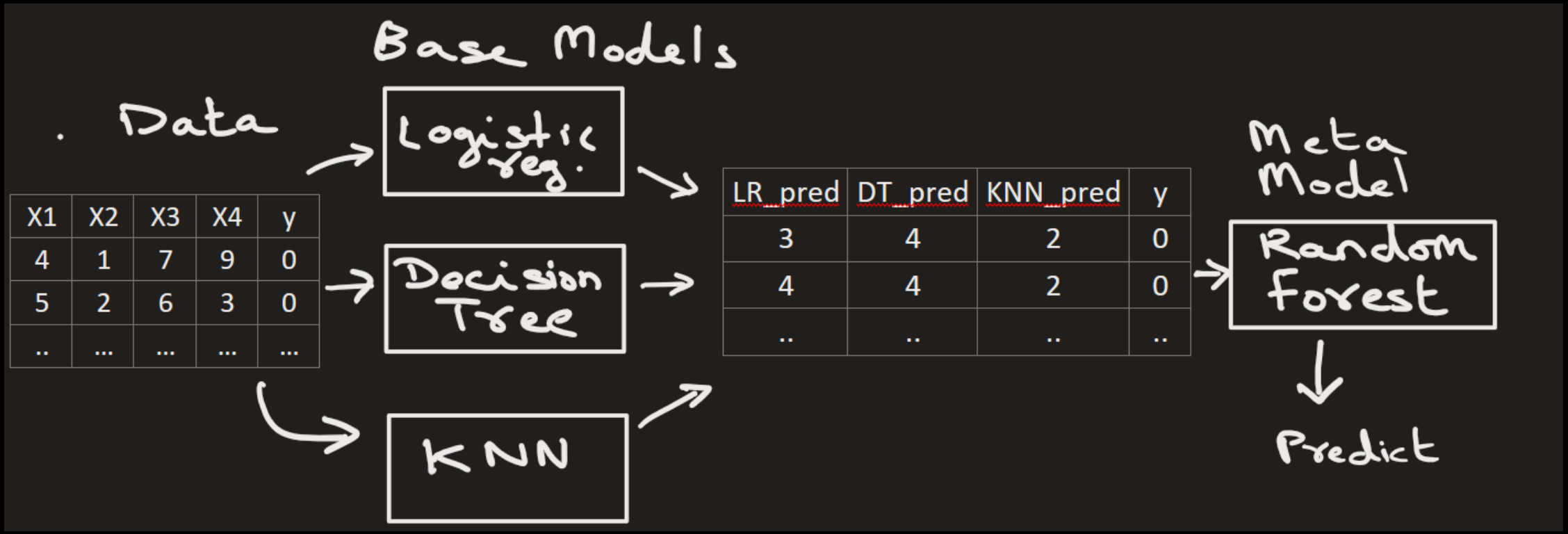
Training multiple models and combining their predictions.

Stacking:

1. Train multiple **base models** (e.g., KNN, Decision Tree, Logistic)
2. Each base model makes predictions
3. These predictions become **inputs** to a new model
4. The **meta-model** (e.g. Random Forest) learns how to combine them



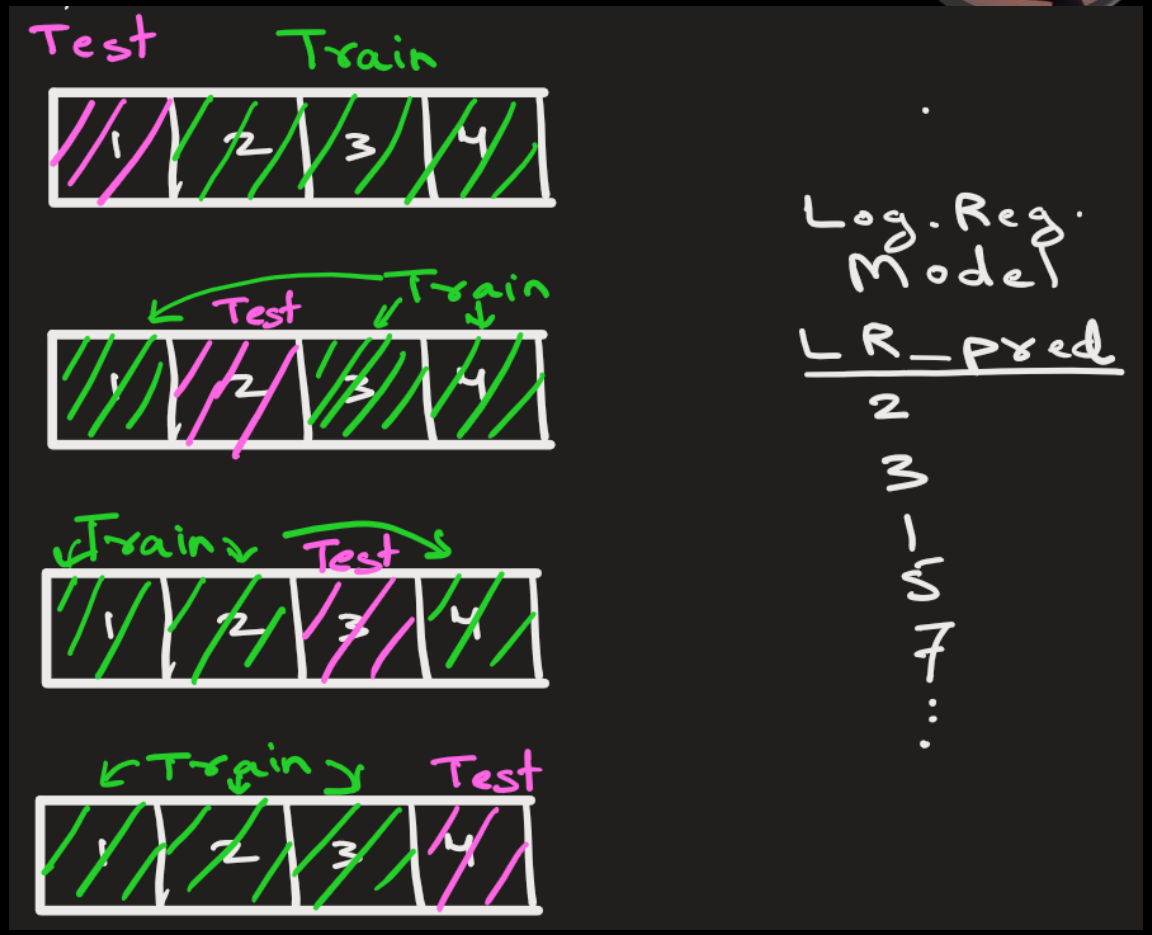
- **Base models should be diverse**, not similar.
- The **meta model** does not see the features X
- The **meta model** sees the prediction y



We perform cross validation when we feed data to base models.
Assume **4-Fold Cross Validation**.

Data Split Example

- Fold 1 → validation
- Fold 2 → validation
- Fold 3 → validation
- Fold 4 → validation



Process for ONE base model (say, Log. Reg.):

- Train on folds
- Predict on remaining fold
- Repeat for all folds on this ONE model

Result:

Every row gets a prediction from a model that **did NOT train on it**.
The meta model does not overfit.

We repeat this process for all base models.

Then we feed the result to the meta model.

LR_pred	DT_pred	KNN_pred	y
3	4	2	0
4	4	2	0
..	..	..	..

→

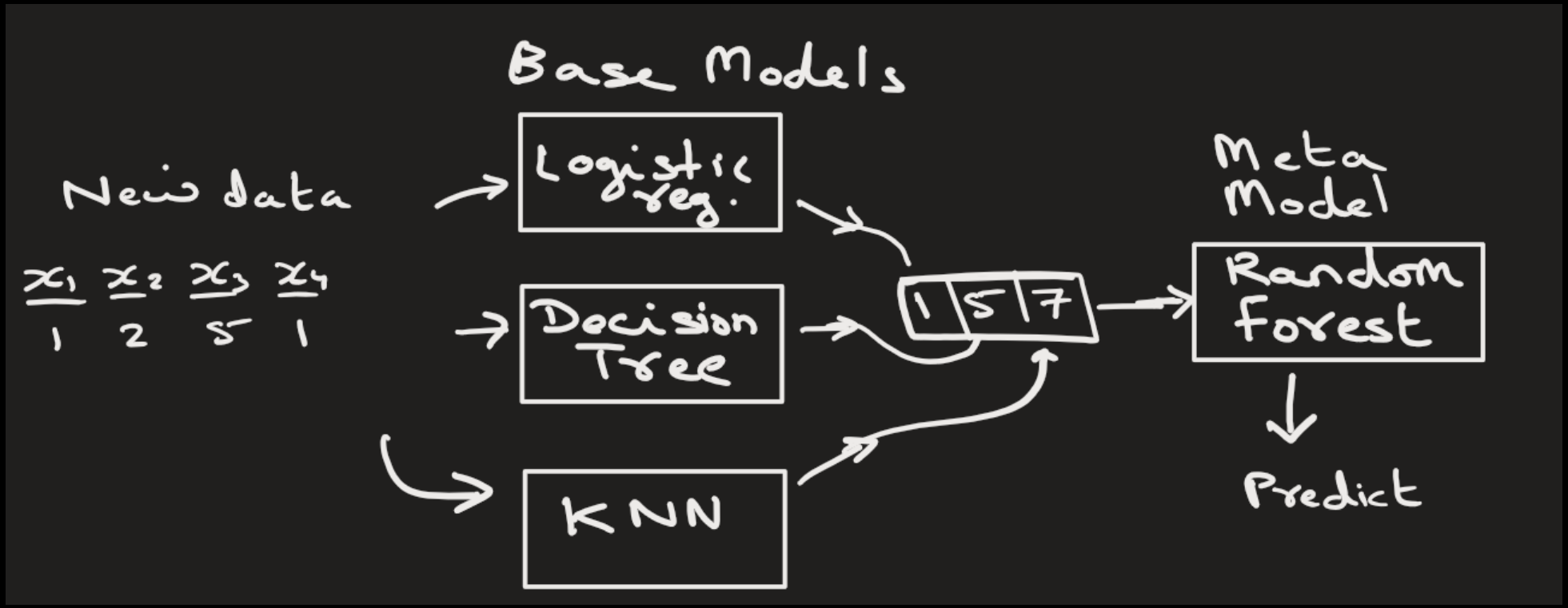
Meta Model

Random Forest

↓

Predict

So, when we get new data and want to make predictions, we feed the data to the base models, which then feeds to the meta model and then it makes predictions.



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